

Checkpoint, 28 September 2026: the BPS uplift decoder of $\mathcal{N}=2$ SYK — verified structure, its variance, and deviation from freeness as nonzero-length wormholes

syk_gravity project

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Abstract

This note records where the project stands after the September 2026 audit and one day of follow-up work. It has two parts.

Part I restates the results about the uplift decoder $O = P_B(1 - n_i)P_B$ of one-flavor $\mathcal{N}=2$ SYK that survived the audit: exact structure, BPS dimensions, the Haar/Wachter *null* model, and robust numerics. It adds new checks: the charge-sector structure of the BPS space, and an exact index formula reproducing every measured BPS dimension. It also summarizes a new numerical campaign that measured the “chord invariants” (free cumulants, crossing weights, six-letter words) of the decoder for $q = 3$, $N' \leq 16$ and $q = 5$, $N' \leq 17$.

Part II is analytic.

- **The variance is a zero-energy two-point function.** The decoder variance is exactly the zero-energy two-point function of the neutral bilinear $n_i - \mu$ in a *single* R-charge sector.
- **The super-Schwarzian gets the structure but not the size.** We re-derive the super-Schwarzian (LMRS) answer from the zero-energy wavefunctions. It reproduces the parity and R-charge structure of the data with no free parameters, but overshoots the magnitude by 15–60 %. We trace this to the absence of a conformal window at $N' \lesssim 16$.
- **The finite- λ chord formula matches.** The double-scaled super-chord formula of Boruch, Lin and Yan [4], evaluated at $\lambda = 2p^2/N'$ with no free parameters, reproduces the measured variance to about 2 % for $N' = 5 \dots 16$.
- **Freeness corresponds to the zero-length wormhole.** In that theory the Haar/Wachter value of the second free cumulant is exactly the contribution of the *zero-length* supersymmetric wormhole. The deviation from freeness is the contribution of wormholes of nonzero length:

$$r = a + \sum_{n \geq 1} P_n q^{2\Delta_n}.$$

Each statement is tagged as [\[literature\]](#), [\[derived\]](#), [\[numerical\]](#), [\[interpretation\]](#) or [\[conjecture\]](#).

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1 Scope, status labels, and notation

Status labels. [literature]: stated in a primary paper in papers/ (cited). [derived]: follows from stated inputs by the derivation shown, or checked here to machine precision. [numerical]: observation from exact numerics at the sizes available. [interpretation] / [conjecture]: reading or hypothesis, not established.

Model. One-flavor $\mathcal{N}=2$ SYK [2, 1]: N' complex fermions and a p -body supercharge $Q = \sum_{i_1 < \dots < i_p} C_{i_1 \dots i_p} \psi^{i_1} \dots \psi^{i_p}$ with i.i.d. complex Gaussian couplings, and $H = \{Q, Q^\dagger\}$. We use $p = q = \hat{q} = 3$ unless stated. In the code, Q acts as $\omega \wedge$ on $\Lambda^P(\mathbb{C}^{N'})$.

- N' is the size of the *enlarged* theory (old size $N = N' - 1$), and $P = \lfloor N'/2 \rfloor$.
- $D = \binom{N'}{p}$ is the dimension of the charge sector. B is the BPS subspace ($\ker Q \cap \ker Q^\dagger$) at charge P , $d = \dim B$, and P_B its projector.
- $a = d/D$ is the BPS fraction. $\tau = \text{Tr}_B / d$.

Decoder. $O = P_B \Pi P_B$ restricted to B , with $\Pi = 1 - n_i$ (the “slot” projector; $i = N' - 1$ is the uplift decoder of [1]). With i.i.d. couplings every mode is statistically equivalent, so we average over several modes.

- Eigenvalues $\lambda \in [0, 1]$ are squared cosines of principal angles between B and the slot.
- UV mean $b = \text{Tr}_D \Pi / D = 1 - P/N'$.
- Central quantity:

$$r \equiv \frac{\text{Var}(\lambda)}{b(1-b)}, \quad \text{Var}(\lambda) = \tau(O^2) - \tau(O)^2. \quad (1)$$

r is the fraction of the “UV” Bernoulli variance that survives BPS projection. $0 \leq r \leq 1$.

Part I — Verified structure and numerics

2 The safe core of the audit

The audit (`docs/project_archaeology.md`, `research/notes/gravitational_dual_fragility_audit.md`) checked the earlier draft claim by claim. What survived:

2.1 Exact structure

1. [literature] $0 \preceq O \preceq 1$, with $\lambda = \cos^2$ of the principal angles; definition as in [1, eqs. 101–102]. The BPS projector has the Hodge form $P_B = 1 - QGQ^\dagger - Q^\dagger GQ$, with G the Green’s function on the non-BPS part.
2. [derived] *Exact atoms.* $\lambda = 1$ eigenvectors are BPS states of the old theory in the “empty” copy that are annihilated by wedging with the new-mode couplings. $\lambda = 0$ eigenvectors are the analogous states in the “occupied” copy. (Characterization in the audit; implemented in `src/protected_atoms.py`.) [numerical] The atom fraction is 55.6% ($N = 8, 9$), 32.9% (10, 11), 5.6% (12, 13) and 0 (14–16). This is a monotone decay close to rank lower bounds, not a resonance.
3. [derived] *Mean and pairing.* The disorder-averaged mean is $\langle \lambda \rangle = b$ (exchangeability of modes plus $\sum_i n_i = P$). Per realization this holds only at even N' , half filling, where particle–hole symmetry pairs $\lambda \leftrightarrow 1 - \lambda$.
4. [derived] *Sum rule.* On the charge- P sector, $\sum_i (n_i - \mu) = 0$. Hence the BPS-projected covariance of two different modes is exactly $c = -1/(N' - 1)$. Measured: $-0.250, -0.200, \dots, -0.067$ for $N' = 5 \dots 15$.

2.2 BPS dimensions and charge sectors

1. [literature] $d = 2 \cdot 3^{N'/2-1}$ (even N' , $P = N'/2$) and $d = 3^{(N'-1)/2}$ (odd N') [1].
2. [derived] + verified. Fermion number fixes the R-charge. With ψ carrying charge $1/p$ and Q charge 1,

$$j = \frac{P - N'/2}{p} \Rightarrow j = 0 \text{ (} N' \text{ even)}, \quad j = -\frac{1}{6} \text{ (} N' \text{ odd)}. \quad (2)$$

So B is a *single* R-charge sector. Exact counts: even N' has BPS states only at $j = 0, \pm\frac{1}{3}$, with dimensions $(54; 27, 27)$, $(162; 81, 81)$, $(486; 243, 243)$ for $N' = 8, 10, 12$. The ratio $d_{\pm 1/3}/d_0 = \frac{1}{2}$ is exactly $\cos(\pi/3)/\cos 0$, as predicted by $\text{Tr } P_j \propto \cos \pi j$ [3, eq. 82]. Odd N' has $j = \pm\frac{1}{6}$.

3. [derived] + verified. The refined-index formula [4, eq. A.4],

$$\frac{d_j}{2^{N'}} = \frac{1}{p} \sum_{n=0}^{p-1} \sin^{N'}\left(\frac{\pi n}{p}\right) e^{i\pi j p} e^{-2\pi i j n}, \quad |j| < \frac{1}{2}, \quad (3)$$

reproduces *every* measured BPS dimension in every sector $|j| < \frac{1}{2}$ for $N' \leq 16$ (index saturation; test `tests/test_dssyk_n2.py`). The inverse Fourier transform resolves j only mod 1, hence the restriction.

2.3 The Haar / Wachter null model

Replace P_B by a Haar-random projector of the same rank. Free probability then predicts that the law of O is the *free compression* of $\text{Bern}(b)$ by a free projection of trace a : the Wachter (MANOVA) law.

- [literature] (Nica–Speicher [5]) Compressing a variable x by a free projection p of trace t gives

$$\kappa_n(pxp) = t^{n-1} \kappa_n(x) \quad (\text{free cumulants, computed in the compressed algebra}). \quad (4)$$

- Consequence [derived]: for the Haar null model, $\rho_n \equiv \kappa_n / \kappa_n^{\text{Bern}(b)} = a^{n-1}$. In particular

$$r_{\text{Haar}} = \rho_2 = a. \quad (5)$$

- [derived] (audit) Exact finite- D Weingarten moments hold, e.g. $m_2 = b[D^2(a+b-ab)-1]/(D^2-1)$, as does the genus-one resolvent $W_1(x) = -b(1-a)(1-b)x(x-1)/[(x-\lambda_-)(x-\lambda_+)]^{5/2}$ (checked through $k = 5$ and 6). These are properties of the *null model*. The audit found that the earlier “gravity = Weingarten genus expansion” reading is not supported.
- [numerical] Our Haar runs reproduce (5) and $\rho_{4,6} = a^{3,5}$ to three digits for $N' \geq 7$. This validates the numerical pipeline.

2.4 Robust numerics

[numerical] Interior level statistics of O are GUE ($\langle r_{\text{gap}} \rangle = 0.599 \pm 0.001$ at $N' = 12$ [1]). The interior density is *not* Wachter: mid-band weight is depleted about $2\times$ and moves to the walls. Of order 10–15% of the spectrum lies below the Wachter edge at $N' = 13$ –16.

2.5 Retracted

From the earlier draft (details in the audit):

- the end-of-world-brane dual with $e^{S_0} = D$;
- “BPS projection trivializes the Schwarzian”;
- “crossing = handle $\propto D^{-2}$ ”;
- the self-energy formula for O ;

- the “tail = image of the near-BPS spectrum” explanation;
- the bordered-Toeplitz closed form.

The Kunisky–Chebyshev error identity [6] holds for *any* pair of projections, so it is bookkeeping, not evidence.

3 Chord invariants of the decoder

What was measured. All quantities are exact linear algebra, per disorder realization (`src/chord_invariants.py`; `research/notes/chord_invariants_2026-09-28.md`). For centered operators $\hat{A}_i = P_B(\Pi_i - \mu)P_B$ with $v_i = \tau(\hat{A}_i^2)$:

- free cumulants κ_n ($n \leq 6$) of O , and the ratios $\rho_n = \kappa_n / \kappa_n^{\text{Bern}(b)}$;
- the crossing weight $X = \tau(\hat{A}_i \hat{A}_j \hat{A}_i \hat{A}_j) / (v_i v_j)$ and the non-crossing factorization $F = \tau(\hat{A}_i \hat{A}_i \hat{A}_j \hat{A}_j) / (v_i v_j)$ for $i \neq j$;
- all six-letter pair words, compared with X^{cr} , where cr is the crossing number.

The fully crossing word $ijkijk$ discriminates between two rules: free compression predicts X^2 and a single-parameter chord (q -Gaussian) rule predicts X^3 . We quote $e_3 = \ln w_3 / \ln X$.

q	N'	model	a	r	ρ_4 (a^3)	X	F	e_3
3	10	SYK	0.643	0.746	0.435 (0.266)	0.810	0.995	2.29 ± 0.01
3	12	SYK	0.526	0.687	0.348 (0.146)	0.785	0.995	2.35 ± 0.01
3	14	SYK	0.425	0.642	0.290 (0.077)	0.769	0.994	2.41 ± 0.01
3	16	SYK	0.340	0.601	0.245 (0.039)	0.755	0.995	2.42 ± 0.03
3	16	Haar	0.340	0.340	0.039	0.346	1.003	1.94
5	17	SYK	0.874	0.908	0.751 (0.668)	0.931	0.998	$w_3/X^2 = 0.979$
5	17	Haar	0.874	0.874	0.668	0.874	1.000	$w_3/X^2 = 1.000$

Table 1: [\[numerical\]](#) Selected chord invariants (mean over realizations; 60→2 realizations per point as N' grows; full tables in `results/data/chord_invariants_2026-09-28/`).

Findings [\[numerical\]](#).

1. **SYK is not free compression, and the gap grows with N' .** r/a rises from 1.04 ($N' = 7$) to 1.77 ($N' = 16$).
2. **The single-operator law is “Wachter with a renormalized parameter” only in its low cumulants.** $\rho_n^{1/(n-1)}$ nearly coincide, but the KS distance of the spectrum to $\text{Wachter}(t = r)$ is 0.15–0.23.
3. **Words with up to two crossings are multiplicative.** $F = 1$ and $\text{cr} \leq 2$ words equal X^{cr} to $\lesssim 1\%$.
4. **The fully crossing word obeys neither rule.** $e_3 \simeq 2.3$ –2.4, drifting slowly upward: neither free (2) nor single-parameter chord (3). At $q = 5$, $X \approx 0.95$ makes e_3 ill-conditioned, but the ratios show the same pattern.
5. **Rigidity grows but decelerates.** $\eta_r = (r - a)/(1 - a)$ grows as 0.29, 0.34, 0.38, 0.40 ($N' = 10$ –16). Section 7 explains this.

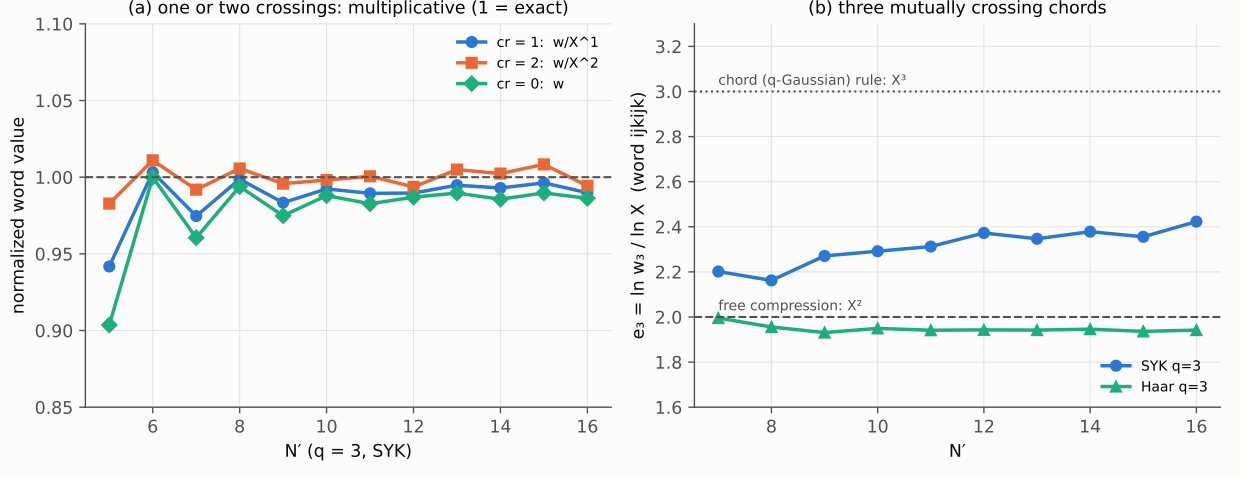


Figure 1: [numerical] Six-letter pair words, $q = 3$ SYK. (a) Words with 0, 1, 2 crossings are multiplicative in X . (b) Effective exponent of the fully crossing word: Haar follows free compression ($e_3 \approx 2$). SYK lies between the free and single-parameter chord rules. Error bars (≈ 0.01) are smaller than the markers.

Part II — The decoder variance and the wormhole

4 The variance is a zero-energy two-point function

Proposition 1 ([derived], exact). *Let $\mu = \tau(P_B n_i P_B)$ and $\mathcal{O} = n_i - \mu$. Then*

$$\text{Var}(\lambda) = \frac{\text{Tr}[P_j \mathcal{O} P_j \mathcal{O}]}{\text{Tr} P_j}, \quad P_j \equiv P_B = \lim_{\beta \rightarrow \infty} e^{-\beta H} \Big|_{\text{charge } P}, \quad (6)$$

with j given by (2).

Proof. Var is invariant under $\Pi \rightarrow 1 - \Pi$, so we may use n_i . Then $\tau((P_B n P_B)^2) - \tau(P_B n P_B)^2 = \tau(P_B n P_B n P_B) - \mu^2 = \tau(P_B \mathcal{O} P_B \mathcal{O} P_B)$, since $P_B^2 = P_B$ and $\tau(P_B \mathcal{O} P_B) = 0$. Fermion number is conserved, so P_B is the zero-energy projector of the single R-charge sector (2). $\square$

So $\text{Var}(\lambda)$ is the “infinitely-long-time” two-point function of $\mathcal{O}$ in the sense of LMRS [3] (their $\hat{\mathcal{O}} = POP$), with two differences:

- it is normalized by the number of BPS states in *one* sector, $\text{Tr} P_j$, not the total N_{BPS} ;
- $\mathcal{O}$ is the *same-site* bilinear $\bar{\psi}_i \psi_i$, not LMRS’s $\psi^k \bar{\psi}_i$ with $i \neq k$.

At leading large N the two bilinears have the same conformal two-point function, $\langle n_i(t) n_i(0) \rangle_c = \langle \bar{\psi}_i(t) \psi_i(0) \rangle \langle \psi_i(t) \bar{\psi}_i(0) \rangle + O(1/N)$. This is a single cross-contraction, so both have $\Delta = 2 \cdot \frac{1}{2q} = \frac{1}{3}$.

[numerical] In our BPS sectors, $\text{Var}_{\text{hop}} / \text{Var}_{n_i} = 1.14, \dots, 1.06$ for $N' = 8 \dots 14$, tending to 1.

5 The super-Schwarzian answer, and why it overshoots

5.1 Derivation from the zero-energy wavefunctions

Imported [literature] from LMRS [3].

- The two-sided zero-energy state $|Z_j\rangle$ of $\mathcal{N}=2$ super-Liouville quantum mechanics has two fermionic components, $g_{1,2} \propto e^{-\ell/2} K_{\frac{1}{2} \mp j}(2e^{-\ell/2})$ (their eq. 22), normalizable for $|j| < \frac{1}{2}$.
- A neutral matter operator of dimension Δ acts on it as multiplication by $e^{-\Delta\ell}$ (their eq. 54).
- The $\beta \rightarrow \infty$ two-sided state in sector j is $A_{z,j}|Z_j\rangle$ (their eqs. 37, 39).

Step 1 [derived]: reduction to a normalized expectation value. $\text{Tr}[P_j \hat{O} P_j \hat{O}] = |A_{z,j}|^2 \langle Z_j | e^{-\Delta\ell} | Z_j \rangle$. At $\Delta = 0$ this must equal $\text{Tr} P_j$, so

$$\frac{\text{Tr}[P_j \hat{O} P_j \hat{O}]}{\text{Tr} P_j} = \frac{\langle Z_j | e^{-\Delta\ell} | Z_j \rangle}{\langle Z_j | Z_j \rangle}. \quad (7)$$

Neither e^{S_0} nor the TFD coefficient enters; only the shape of the wavefunction does.

Step 2 [derived]: a Mellin integral. For a neutral operator the $U(1)$ integral is trivial. With $x = 2e^{-\ell/2}$ ($d\ell = -2dx/x$, $e^{-\ell} = x^2/4$, $e^{-\Delta\ell} = (x/2)^{2\Delta}$),

$$I_\nu(\Delta) \equiv \int_{-\infty}^{\infty} d\ell e^{-(1+\Delta)\ell} K_\nu(2e^{-\ell/2})^2 = 2^{-1-2\Delta} \int_0^\infty x^{1+2\Delta} K_\nu(x)^2 dx.$$

Use the Mellin transform (Gradshteyn–Ryzhik 6.576.4 with equal arguments [7]; checked numerically to 10^{-11}):

$$\int_0^\infty x^{s-1} K_\nu(x)^2 dx = \frac{\sqrt{\pi} \Gamma(\frac{s}{2}) \Gamma(\frac{s}{2} - \nu) \Gamma(\frac{s}{2} + \nu)}{4 \Gamma(\frac{s+1}{2})}, \quad \text{Re } s > 2|\nu|.$$

Set $s = 2 + 2\Delta$ and sum the two components $\nu = \frac{1}{2} \mp j$. Using $\Gamma(z+1) = z\Gamma(z)$ on the second factors,

$$\Gamma(\Delta + \frac{1}{2} + j) \Gamma(\Delta + \frac{3}{2} - j) + \Gamma(\Delta + \frac{1}{2} - j) \Gamma(\Delta + \frac{3}{2} + j) = (2\Delta + 1) \Gamma(\Delta + \frac{1}{2} + j) \Gamma(\Delta + \frac{1}{2} - j).$$

Then $(2\Delta + 1)/\Gamma(\Delta + \frac{3}{2}) = 2/\Gamma(\Delta + \frac{1}{2})$, and Legendre duplication, $\Gamma(\Delta) \Gamma(\Delta + \frac{1}{2}) = 2^{1-2\Delta} \sqrt{\pi} \Gamma(2\Delta)$, give

$$\mathcal{N}_j(\Delta) \equiv I_{\frac{1}{2}-j} + I_{\frac{1}{2}+j} = \frac{1}{8} \frac{\Delta \Gamma(\Delta)^2 \Gamma(\Delta + \frac{1}{2} + j) \Gamma(\Delta + \frac{1}{2} - j)}{\Gamma(2\Delta)}. \quad (8)$$

The convergence condition at $\Delta = 0$ is exactly $|j| < \frac{1}{2}$, the normalizability condition of $|Z_j\rangle$.

Step 3 [derived]: normalize. $\mathcal{N}_j(0) = \frac{1}{4} \Gamma(\frac{1}{2} + j) \Gamma(\frac{1}{2} - j) = \pi/(4 \cos \pi j)$. This reproduces the norm $\langle Z_j | Z_j \rangle \propto 1/\cos \pi j$ of [3, eq. 26], a check on Step 2. Hence

$$\langle e^{-\Delta\ell} \rangle_j = \frac{\cos \pi j}{2\pi} \frac{\Delta \Gamma(\Delta)^2 \Gamma(\Delta + \frac{1}{2} \pm j)}{\Gamma(2\Delta)} \equiv F(\Delta, j), \quad (9)$$

which is exactly LMRS eq. (66) divided by $\text{Tr} P_j = e^{S_0} \cos \pi j$.

Step 4 [literature]: SYK normalization. Converting from Schwarzian to SYK units multiplies by $b_q^2(2CJ)^{-2\Delta}$, with $b_q = [\tan(\pi/2\hat{q})/2\pi]^{1/\hat{q}}$ and $C = \alpha_S N/J$, $\alpha_S = 0.00842$ for $\hat{q} = 3$ (LMRS eqs. 64, 78–80). Therefore

$$r_{\text{LMRS}}(N') = \frac{b_q^2 (2\alpha_S N')^{-2/3} F(\frac{1}{3}, j)}{b(1-b)} \quad (10)$$

with no free parameter.

- Checks: the code reproduces the LMRS Table 1 Schwarzian column (0.1112 vs 0.111; 0.0874 vs 0.0874); $F(0, j) = 1$; J cancels.
- Note: LMRS eq. (77) prints $\langle C\bar{C} \rangle = 2J/N$. FGMS [2, eq. 5.4] has $2J/N^2$, which is required for extensivity. This matters only for gap-based checks.

5.2 Comparison

N'	9	10	11	12	13	14	15	16
r (SYK)	0.728	0.746	0.666	0.687	0.617	0.642	0.577	0.601
r_{LMRS}	0.957	0.957	0.833	0.847	0.744	0.765	0.675	0.700
Haar $r = a$	0.643	0.643	0.526	0.526	0.425	0.425	0.340	0.340

Table 2: [numerical] Decoder variance ratio against the parameter-free super-Schwarzian prediction (10).

What works [numerical].

- **Parity.** The j -factor $F(\frac{1}{3}, -\frac{1}{6})/F(\frac{1}{3}, 0) = \cos \frac{\pi}{6} \Gamma(1)\Gamma(\frac{2}{3})/\Gamma(\frac{5}{6})^2 = 0.920$ removes the even/odd alternation of r with no parameters.
- **A pre-registered test** at fixed even N' , comparing the $j = \pm \frac{1}{3}$ sectors ($P = N'/2 \pm 1$) with $j = 0$. Predicted $\text{Var}_{1/3} / \text{Var}_0 = \frac{1}{2} \Gamma(\frac{7}{6})\Gamma(\frac{1}{2})/\Gamma(\frac{5}{6})^2 = 0.645$. Measured: 0.642, 0.668, 0.677, 0.683 ($N' = 8 \dots 14$). The Haar value is $0.586 \rightarrow 0.560$, and “no j -dependence” gives 1. So this is a 6% success, although the measured ratio drifts away from the prediction.

What fails [numerical]. The magnitude is too high by 15–60%. The N' -dependence also does not match: the pure $N'^{-2/3}$ shape with free normalization gives $\chi^2/\text{dof} = 335$. A fitted $1 - 2.2/N'$ factor restores agreement, but so does a plain power $N'^{-0.44}$. That fitted factor is not a prediction.

5.3 Diagnosis: no conformal window [derived]

By Steps 1–4, the prediction is the zero-energy average of the conformal matter correlator $b_q^2(Jt)^{-2\Delta}$ with $Jt(\ell) = 2\alpha_S N' e^{\ell/2}$. The weight $\rho_j(\ell) e^{-\Delta\ell}$ fixes which physical times dominate:

N'	10	15	20	50	120
median Jt	0.50	0.75	1.0	2.5	6.0
weight where conformal value > UV maximum $\frac{1}{4}$	67%	49%	35%	4%	0

At our sizes the “zero-energy” correlator is dominated by times *shorter than* $1/J$. There the conformal input overshoots the true correlator, which is bounded by its equal-time value. The effective small parameter is $1/(2\alpha_S N') \approx 60/N'$, and a clean window needs $N' \sim 10^2$. [numerical] Consistent with this: the BPS gap is $0.19J$ at $N' = 14$ (not $\ll J$), and $r_{\text{LMRS}} > 1$ for $N' \leq 8$. A parameter-free toy that caps the conformal correlator at its UV value brings the prediction within 0–10% of the data, for two different operators.

6 The finite- λ super-chord answer

The double-scaled theory ($N, p \rightarrow \infty$ with $\lambda = 2p^2/N$ fixed, $q = e^{-\lambda}$) is solvable by chord methods and is UV-complete. The super-Schwarzian is its triple-scaling limit $\lambda \rightarrow 0$ [4]. Boruch, Lin and Yan construct the two-sided super-chord Hilbert space (chord number n = discrete wormhole length, $\ell = 2\lambda n$) and its zero-energy Hartle–Hawking state $|\Psi, j\rangle$ ([literature], their eqs. 3.1–3.11). They compute the normalized zero-temperature two-point function of a neutral operator ([literature], their eqs. 4.5–4.8):

$$\frac{\langle \Psi, j | q^{2\Delta n} | \Psi, j \rangle}{\langle \Psi, j | \Psi, j \rangle} = \frac{(q^{2+4\Delta}; q^2)_\infty (q^{1\pm 2j}; q^2)_\infty (q^2; q^2)_\infty}{(q^{2+2\Delta}; q^2)_\infty^2 (q^{1+2\Delta\pm 2j}; q^2)_\infty}. \quad (11)$$

As $\lambda \rightarrow 0$ this tends to $(2\lambda)^{2\Delta} F(\Delta, j)$, i.e. to (9) (checked: ratio $1 - 3 \times 10^{-3}$ at $\lambda = 0.3$).

Identification with r [derived]. BLY normalize the matter operator so that $W_L W_R \propto q^{\Delta(n_O + n_X)}$ with unit constant, i.e. its infinite-temperature two-point function in the charge sector equals 1. Since $|\Psi, j\rangle \propto \Pi_{0,j} |\Omega, j\rangle$, eq. (11) equals $[\text{Tr}(P_j W P_j W) / \text{Tr} P_j] / [\text{Tr}_j(WW) / \text{Tr}_j \mathbb{K}]$. For $W = n_i - \mu$ on Λ^P , the denominator is exactly $b(1 - b)$. By Proposition 1,

$$r = \text{eq. (11)} \Big|_{\lambda=2p^2/N', \Delta=1/p, j=(P-N'/2)/p} \quad (12)$$

Here $\Delta = 1/p$ because $\bar{\psi}_i \psi_i$ has one ψ and one $\bar{\psi}$ (BLY eq. 1.8). No conformal normalization and no α_S enter.

N'	j	λ	r (SYK)	r chord	r_{LMRS}	SYK/chord	a exact	a chord
6	0	3.00	0.905	0.913	1.345	0.990	0.900	0.900
8	0	2.25	0.819	0.832	1.110	0.984	0.771	0.789
10	0	1.80	0.746	0.756	0.957	0.987	0.643	0.671
12	0	1.50	0.687	0.690	0.847	0.996	0.526	0.559
13	$-\frac{1}{6}$	1.38	0.617	0.613	0.744	1.007	0.425	0.456
14	0	1.29	0.642	0.635	0.765	1.011	0.425	0.459
15	$-\frac{1}{6}$	1.20	0.577	0.565	0.675	1.021	0.340	0.371
16	0	1.13	0.601 ± 0.005	0.589	0.700	1.022	0.340	0.373
14	$-\frac{1}{3}$	1.29	0.449	0.418	0.504	1.074	0.243	0.265

Table 3: [numerical] Exact $q = 3$ SYK against the parameter-free finite- λ prediction (12). The BPS fraction is from the exact index (3) and from the double-scaled formula $D(j)$ [4, eq. 3.17].

Results [numerical].

- r agrees within 2.2% for every $N' = 5 \dots 16$, including parity, with no rescaling (Fig. 2). The super-Schwarzian misses by 15–60% over the same range.

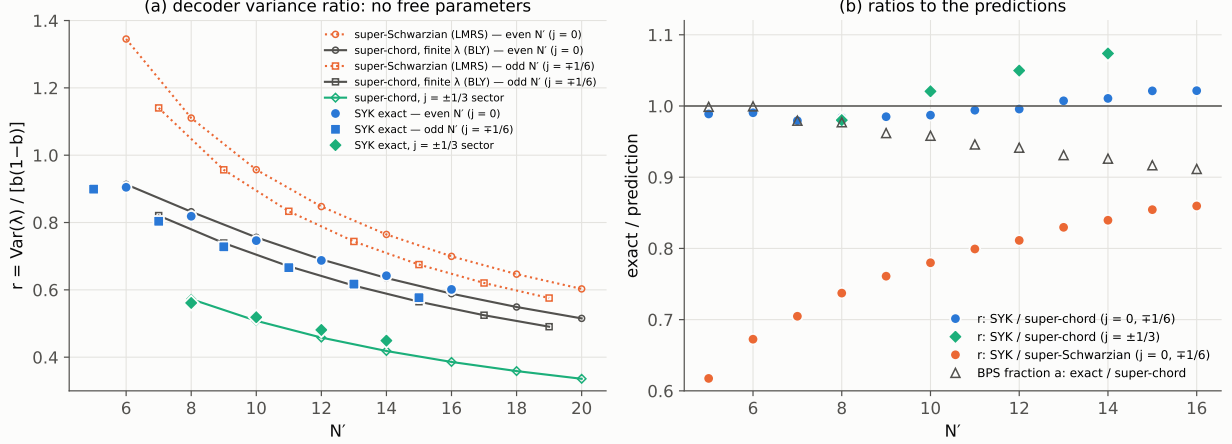


Figure 2: [numerical] (a) r versus N' : exact SYK (filled), finite- λ super-chord (solid; eq. 12), super-Schwarzian (dotted; eq. 10); no free parameters. (b) Ratios to the predictions, including the BPS fraction a .

- There is a small systematic drift in SYK/chord ($0.98 \rightarrow 1.02$). In the $j = \pm \frac{1}{3}$ sectors the agreement degrades to +7% at $N' = 14$.
- The double-scaled BPS fraction overshoots the exact one by up to 10%. This is a pure finite- p effect: $\cos^{N'}(\pi/2p)$ versus $e^{-N'\pi^2/8p^2}$.

Caveats.

- The λ identification is BLY's standard $\lambda = 2p^2/N$, fixed before comparing. The prediction scales as $r \propto \lambda^{0.4-0.6}$, so a 5% ambiguity in λ moves r by about 2.5%. The 2% agreement is at the resolution of the double-scaled mapping at $p = 3$.
- The double-scaled theory at $\lambda = 18/N'$ is not the $N' \rightarrow \infty$ limit at fixed $p = 3$. Its Schwarzian coupling is $\alpha_S = 1/(8p^2) = 0.0139$ versus 0.00842. Agreement must eventually degrade at large N' ; the positive drift may be its onset.

Interpretation [interpretation]. Together with the diagnosis above: at accessible sizes the decoder variance sits in a *double-scaled* regime ($\lambda \approx 1.1-3.6$), not a Schwarzian one. The chord theory, as the UV completion, captures the crossover the Schwarzian misses. BLY draw the same lesson for the ground-state degeneracy itself.

7 Deviation from freeness = wormholes of nonzero length

The project's framing asks for “the deviation from freeness” of the decoder law. The chord theory gives this deviation a precise geometric meaning, at least for the second free cumulant.

7.1 Two facts about the same distribution

Let P_n be the probability that the supersymmetric wormhole $|\Psi, j\rangle$ has chord number n . It is the n -sector weight of $|\Psi, j\rangle$, normalized by eq. 3.11 of [4]; different n are orthogonal.

Fact 1: the BPS fraction is the zero-length probability [literature] [4, eqs. 1.1, 3.12–3.15]. Slice the fixed-charge partition function two-sidedly: $Z(j, \beta)/Z(j, 0) = \langle \Omega, j | e^{-\beta H} | \Omega, j \rangle / \langle \Omega, j | \Omega, j \rangle$, where $|\Omega, j\rangle$ is the empty ($n = 0$) wormhole. As $\beta \rightarrow \infty$, $e^{-\beta H}$ projects onto the chord ground state:

$$a = \frac{d}{D} = \frac{Z(j, \infty)}{Z(j, 0)} = \frac{|\langle \Omega, j | \Psi, j \rangle|^2}{\langle \Omega, j | \Omega, j \rangle \langle \Psi, j | \Psi, j \rangle} = P_0. \quad (13)$$

Fact 2: r is the propagator averaged over wormhole lengths [derived]. By (12) and $W_L W_R = q^{2\Delta n}$ on empty wormholes (a scalar on each n -sector),

$$r = \sum_{n \geq 0} P_n q^{2\Delta n}. \quad (14)$$

7.2 The identity

Since $q^{2\Delta \cdot 0} = 1$, facts (13)–(14) combine into

$$r = \underbrace{a}_{\text{Haar / Wachter, eq. (5)}} + \sum_{n \geq 1} P_n q^{2\Delta n}, \quad \eta_r \equiv \frac{r - a}{1 - a} = \langle q^{2\Delta n} \rangle_{n \geq 1} \quad (15)$$

[derived] Verified from the Hartle–Hawking recursion (BLY eqs. 3.2–3.11) for $N' = 8, 12, 14, 16$: $\sum_n P_n = 1$, $P_0 = D(j)$, and $\sum_n P_n q^{2n/3}$ equals (11) to 10^{-10} or better (hh_length_distribution, test test_free_compression_is_zero_length_term).

Reading [interpretation].

- The free-probability value of the second free cumulant, $\rho_2 = a$, is *exactly* the contribution of the zero-length wormhole.
- The deviation from freeness, $r - a \geq 0$, is the contribution of wormholes of nonzero length, each weighted by the matter propagator $q^{2\Delta n} = e^{-\Delta \ell}$.
- The rigidity fraction η_r is the propagator averaged over nonzero-length wormholes.
- In particular $r \geq a$ is automatic in the chord theory. [numerical] The realization-averaged r satisfies it within errors at every (q, N') measured, and by many standard errors for $q = 3$, $N' \geq 7$ (e.g. $r/a = 1.77$ at $N' = 16$). Individual realizations violate it only where $a \gtrsim 0.9$ and d is tiny ($q = 3$, $N' = 5, 6$; $q = 5$, $N' = 9, 10$; 2 of 300 at $q = 3$, $N' = 7$). There $r \approx a$ and sample fluctuations cross; the minimum single-realization r/a is 0.965.

Figure 3(a) shows the decomposition at $N' = 14$: $a = P_0 = 0.459$ of $r = 0.635$ comes from $n = 0$, and the remaining 0.176 from $n = 1, 2, 3, \dots$

7.3 Consequences

1. η_r **turns over** [numerical]/[interpretation]. In the chord theory η_r *peaks* near $N' \approx 22$ ($\eta_r \approx 0.36$) and then decays. The measured deceleration ($0.34 \rightarrow 0.38 \rightarrow 0.40$ at $N' = 12, 14, 16$) is the approach to this maximum, not a plateau. This is a sharp, testable prediction for $N' \gtrsim 20$ (stochastic trace estimation). The turnover is robust beyond the double-scaled approximation: at fixed $p = 3$, LMRS also gives $r \propto N'^{-2/3} \rightarrow 0$.

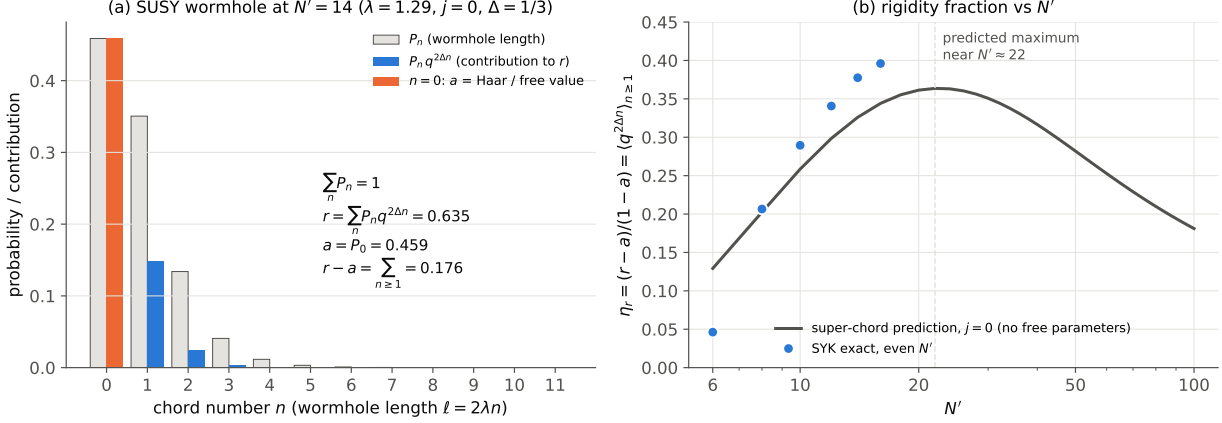


Figure 3: (a) [derived] Wormhole-length distribution P_n of the supersymmetric Hartle–Hawking state at $N' = 14$ ($\lambda = 18/14$, $j = 0$) and its contribution $P_n q^{2\Delta n}$ to r . The $n = 0$ bar is the BPS fraction a , the entire Haar/free value. (b) [numerical] vs prediction: η_r from the chord theory (line; no parameters) and exact SYK (points). The chord curve peaks near $N' \approx 22$. The measured values lie about 0.04 above it, consistent with the finite- p excess of a_{chord} over the exact a . At $N' = 6$, $1 - a = 0.1$, so η_r is poorly conditioned.

2. **Q1 settled structurally.** $a \sim e^{-\pi^2 N'/72}$ in the chord theory (exactly, at $p = 3$, $a \propto \cos^{N'}(\pi/6)$ up to the binomial normalization) is exponentially small, while $r \sim \lambda^{2\Delta} \propto N'^{-2/3}$ is power-law small. So $r/a \rightarrow \infty$: the decoder law moves ever further from Wachter, while $r \rightarrow 0$ (ever further from the commuting UV).
3. **Conjecture [conjecture].** The *entire* free-compression (Wachter/MANOVA) law is the zero-length truncation of the super-chord computation of the decoder moments. Only m_2 has been tested.

8 Open problems and next steps

1. **Zero-temperature OTOC: X and e_3 .** BLY §5 constructs one-particle wormholes and the operator $W_L W_R \sim q^{\Delta(n_O + n_X)_{\text{tot}}}$ (their eq. 5.18), but does not evaluate the zero-temperature OTOC. For our operators the matter–matter crossing weight must be site-local: 1 for disjoint sites, since those bilinears commute, rather than the random-operator q^{Δ^2} . This gives a parameter-free prediction of $X \approx 0.73$ – 0.85 and $e_3 \approx 2.3$ – 2.4 (Table 1).
2. **Higher moments and the density.** $m_k = \tau(O^k)$ are zero-temperature k -point functions of *one fixed projector*. Writing $n_i = \bar{\psi}_i \psi_i$ with ψ_i a single-fermion chiral chord restores a Wick structure, because the infinite-temperature state is Gaussian. Same-site fermion chords then cross with a sign. Testing the conjecture above requires these.
3. **Finite- p corrections.** Explain the $\pm 2\%$ drift and the $j = \pm \frac{1}{3}$ deviation. Candidates: exact finite- N intersection weights (e.g. $1 - 2p/N$ vs $e^{-2p/N}$ for a single-site bilinear crossing a 3-body chord), and the λ identification.
4. **Larger N' .** Test the η_r maximum near $N' \approx 22$ and the breakdown of double scaling.
5. **Framing.** The project overview’s “planar chords = non-crossing partitions (Narayana)

$\Rightarrow$ Wachter” does not describe the actual $\mathcal{N}=2$ chord rules. Those have directed chords, friend/enemy weights and q^Δ matter weights, and give Wachter only as the zero-length piece.

A Reproducibility

All numbers are exact linear algebra (no stochastic trace estimation), with the same coupling generator and seeds as `src/decoder_fast.py`. Code state: commit 081877d plus the working-tree files added for this note (`src/dssyk_n2.py` [hh_length_distribution], `scripts/plot_wormhole_freeness.py`).

Item	Code / data
Chord invariants, Tab. 1, Fig. 1	<code>scripts/run_chord_invariants.py</code> , <code>scripts/chord_invariants_campaign.sh</code> (groups A–F2), <code>scripts/analyze_chord_invariants.py</code> ; <code>results/data/chord_invariants_2026-09-28/</code> (JSONL per realization, <code>meta_*.json</code> with args and commit)
Super-Schwarzian, Tab. 2	<code>src/lmrs_predictions.py</code> , <code>scripts/compare_lmrs_variance.py</code> , <code>scripts/lmrs_mismatch_checks.py</code> , <code>scripts/lmrs_jsector_test.py</code> ; <code>results/data/lmrs_variance_2026-09-28/</code>
Super-chord, Tab. 3, Figs. 2–3	<code>src/dssyk_n2.py</code> , <code>scripts/compare_dssyk.py</code> , <code>scripts/plot_wormhole_freeness.py</code> ; <code>results/data/dssyk_2026-09-28/</code>
Tests	<code>tests/test_chord_invariants.py</code> , <code>tests/test_lmrs_predictions.py</code> , <code>tests/test_dssyk_n2.py</code>
Derivations (living)	<code>docs/derivations.md</code> D1, D2

Realizations. $q = 3$: 60/30 (SYK/Haar) at $N' = 5$ –8; 40/20 at 9–10; 24/12 at 11; 16/8 at 12; 10/5 at 13; 4/2 at 14; 3/2 at 15; 2/1 at 16. $q = 5$: 20/10 at 9–14; 8/4 at 15; 4/2 at 16; 2/1 at 17. j -sector test: seeds 0–3, $N' = 8$ –14.

Memory safety. The two largest runs ($q = 3$, $N' = 16$; $q = 5$, $N' = 17$) were OOM-killed once when run concurrently. They were rerun after two changes: a partial eigendecomposition of $\{Q, Q^\dagger\}$ with a certified spectral gap, and a byte-bounded cache. Regression against stored rows: $\leq 4 \times 10^{-16}$. All heavy jobs now run under `scripts/memwatch.py` (per-job cap, 30 GB project budget). Peaks were 7.5 and 7.0 GB.

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